

A Guide to the WC 2026 Prediction Pages

What every number, chart, and indicator means — across all three pages.

There are three pages in the WC 2026 prediction section. Each draws from the same underlying joint score PMF engine. This guide explains every element visible on each page.

Page 1 — Pre-Game Predictions

URL: sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world-cup/pre-match.html

This page lists every World Cup match scheduled for the current day, with the model's pre-game probability estimates and edge analysis. Data loads automatically and refreshes every five minutes from the latest prediction JSON. A red banner appears if the underlying data is more than four hours old, indicating a potential pipeline issue.

KPI Cards

Matches Today: The number of World Cup matches with regulation kickoffs scheduled for today.

Value Bets: The count of individual betting markets across all today's matches where the model's edge is at least 4% and the 90% confidence interval lower bound still exceeds the market's no-vig probability. This is a strict filter; the number is frequently zero or very small.

Best Edge: The single largest edge found across all today's markets, expressed as a percentage. The sub-label identifies the specific market and match. A positive edge means the model assigns higher probability to the outcome than the market does. This does not guarantee a profitable bet.

Avg xG / Match: The average of (λ_{home} + λ_{away}) across today's matches — the model's expected total goals per match. Derived from the composite prior after market reconciliation.

Bankroll Sizing Tool

Enter a bankroll amount and select a Kelly fraction. The table will then display a recommended dollar stake for every flagged value bet. Three fractions are available:

Full Kelly: The mathematically optimal fraction under the Kelly criterion. Produces large drawdowns in practice when model edge estimates contain error. Not recommended unless confidence in the edge estimate is high.

Half Kelly: Bet size divided by two. Substantially reduces variance while retaining most of the theoretical edge. Used as the default in the engine's internal sizing calculations.

Quarter Kelly: Conservative setting. Appropriate when acknowledging significant uncertainty in the model's probability estimates.

The Match Table

Each row represents one match. Columns from left to right:

Match: Home team vs away team. All matches are treated as neutral venue except USA, Canada, and Mexico matches, which carry a small host-advantage adjustment (+0.10 attack lambda, -0.10 defense lambda).

1X2 Probability Bars: Three colored segments proportional to the regulation-time probabilities: Home Win (gold), Draw (gray), Away Win (blue). These are derived by summing the appropriate cells of the joint PMF: home win = sum of all cells where home goals > away goals; draw = main diagonal; away win = all cells where away goals > home goals.

O/U 2.5: Probability that total regulation-time goals exceed 2.5 — i.e., that three or more goals are scored. Computed by summing all PMF cells where home_goals + away_goals ≥ 3.

BTTS: Both Teams To Score — the probability that both teams score at least one regulation-time goal. Computed by summing all cells where home_goals ≥ 1 and away_goals ≥ 1.

Top Score: The single most probable final score and its probability, directly read from the peak cell of the joint PMF grid.

xG (H-A): The model's expected goals for home and away separately (λ_{home} , λ_{away}). These are the Poisson mean parameters after market reconciliation.

Best Edge / Fair Odds: The highest-edge market for this match that passes all three value filters (≥4% edge, CI check, ≥2% liquidity). Fair decimal odds = 1 / model_probability. If no market passes the filters, this cell is blank.

Expanded Row

Clicking a row reveals three sections:

Full Scoreline Distribution: All scored probabilities from 0–0 through the grid maximum, ranked by probability. The top 15 are displayed with proportion bars. These values are read directly from the joint PMF grid with no further transformation.

All Markets: Every market the model has priced: 1X2, Draw No Bet, Double Chance, Over/Under at 0.5 / 1.5 / 2.5 / 3.5 / 4.5, BTTS, and team-level totals (Home Over 0.5 / 1.5, Away Over 0.5 / 1.5). All computed from the same joint PMF.

Edge Report: For each market: Model Probability, Market Implied (no-vig), and Edge%. Rows highlighted in gold pass all three value filters. The Kelly stake column reflects the bankroll and fraction selected above the table.

Page 2 — PMF Distributions

URL: [sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world-cup/pre-match/Probability Distributions.html](https://sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world-cup/pre-match/Probability%20Distributions.html)

This page visualizes the complete probability distribution for every match. Select a match using the navigation chips at the top. All charts update immediately on selection and draw from the same joint PMF as Page 1.

Chart 1 — Joint Score PMF Heatmap

A color-coded grid where each cell (h, a) shows $P(\text{Home} = h, \text{Away} = a)$. Home goals are on the vertical axis; away goals on the horizontal axis.

Color scale: dark blue/black = near zero probability; warm orange/yellow = moderate (3–8%); bright green = the highest-probability cells (typically > 8%). The color mapping uses the square root of the probability relative to the maximum cell, which visually separates low-probability cells that would otherwise appear uniformly dark on a linear scale.

Any market can be read directly from this grid. For example: over 2.5 is the sum of all cells in the top-right region where $h + a \geq 3$; a clean sheet for the home team is the sum of the entire first column ($a = 0$).

The **tail mass** shown below the heatmap is the probability captured in the explicit overflow bucket — scores beyond the grid boundary. The grid values plus the tail mass sum to exactly 100%.

Chart 2 — Marginal Goal Distributions

Two bar charts: one for the home team, one for the away team. Each bar represents $P(\text{team scores exactly } k \text{ goals})$ for $k = 0, 1, 2, \dots$. The marginal for the home team is obtained by summing the joint PMF across all away goal counts for each value of home goals:

$$P(\text{Home} = h) = \sum_a P(\text{Home} = h, \text{Away} = a)$$

The tallest bar for each team is typically at 1 goal for most World Cup group matches. A very tall $P(0)$ bar indicates the model expects the team to be frequently shut out.

Chart 3 — Total Goals Distribution

A bar chart showing $P(\text{total regulation goals} = k)$ for $k = 0, 1, 2, \dots$. Computed by summing all joint PMF cells along each diagonal ($h + a = k$). Common over/under lines (0.5, 1.5, 2.5, 3.5, 4.5) are shown as vertical dividers. The probability to the right of any divider is the Over probability for that line; to the left is the Under probability.

Chart 4 — Goal Difference Distribution

A bar chart centered on zero showing the probability of each possible goal difference (home goals minus away goals). Gold bars indicate home wins (positive difference), gray indicates a draw (zero), blue indicates away wins (negative difference). This chart makes it easy to see whether the model expects a close match or a one-sided one.

Chart 5 — Top 15 Most Likely Scorelines

A ranked bar chart of the 15 highest-probability individual final scores with exact percentages. These are the raw joint PMF values for the 15 cells, sorted in descending order.

For correct score betting: if the market is offering decimal odds of D on score S , the no-vig market-implied probability is approximately $1/D$ adjusted for margin. The edge on that score is $(\text{model_probability} - 1/(D \times \text{margin})) / (1/(D \times \text{margin}))$.

O/U Lines Table

A table listing Over and Under probabilities for every standard total line (0.5, 1.5, 2.5, 3.5, 4.5, 5.5). All computed by summing the appropriate region of the joint PMF.

Page 3 — Live In-Play PMF

URL: [sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world-cup/live/Probability Distributions.html](https://sportsodds.wizardofodds.com/tools/odds-scanner/predictions/world-cup/live/Probability%20Distributions.html)

This page activates when a World Cup match is in progress. When no matches are live it displays 'No live matches right now' along with the next scheduled kickoff. All probabilities remain regulation-time only.

How the Live Model Differs From Pre-Game

The pre-game model asks: what will the final score be from kickoff? The live model asks: given that the current score is H–A at minute t , what will the final score be at 90+ minutes?

This is a conditional probability calculation. The live model only assigns probability to scores reachable from the current state. A match at 2–1 in the 70th minute cannot finish 1–0 or 0–0; those cells of the PMF grid are zero.

The model computes expected remaining goals for each team using a **non-homogeneous hazard model** — meaning the goal rate varies by match minute. The temporal baseline is calibrated from the minute-by-minute goal distribution across the 2018 and 2022 World Cups (128 matches, approximately 330 goals):

- Minutes 1–10: below-average rate (teams settling in)
- Minutes 25–45: rising rate as play opens up
- Minutes 45–50: elevated rate immediately following half-time kickoff
- Minutes 80–90+: above-average rate, particularly when a team is chasing the match

On top of the temporal baseline, **score-state multipliers** scale each team's goal rate based on the current score. These multipliers are calibrated from World Cup data and the football forecasting literature (Dixon & Robinson 1998):

Score state	Effect on goal rates
Draw at minute 60+	Both teams $\times 1.10$ (games open up in final 30 minutes)
Home team losing by 1	Home $\times 1.25$, Away $\times 1.05$ (counter-attack risk)
Home team losing by 2+	Home $\times 1.40$, Away $\times 1.10$
Home team winning by 1	Home $\times 0.90$, Away $\times 1.10$ (away team pushes)
Home team winning by 2+	Home $\times 0.80$, Away $\times 1.15$

When live expected goals (xG) data is available from BallDontLie, the model blends the live xG rate (60% weight) with the pre-game expected rate (40% weight) to produce a more responsive intensity estimate.

Connection Badge

A small indicator in the header shows the data delivery mode:

- **WebSocket (green):** The browser has an active connection to the live prediction server. When BallDontLie sends a goal event or status change, the server recomputes the PMF and pushes it to connected browsers typically within 200 milliseconds. No page refresh is required.
- **Polling (yellow):** The real-time connection is unavailable. The page fetches updated data from a static JSON file every 60 seconds. Updates arrive with up to a one-minute delay. This is the fallback mode.

Live KPI Cards

Matches Live: Number of World Cup matches currently in progress per BallDontLie.

Next Kickoff: Next match starting today, with kickoff time in Eastern time.

Goals Today: Sum of goals scored across all live matches as of the last snapshot.

Data Age: Time elapsed since the last live snapshot was generated. Under two minutes is normal during live matches. A value over ten minutes during a live match triggers the health warning banner.

Win Probability Bar

The same three-segment bar as on Page 1, but now conditional on the current live score and minute. These probabilities update with every new snapshot. They represent the probability of each regulation-time outcome given that the match is currently at score H–A at minute t .

Directly below the bar, a **Pre-game** → **Live Shift** table shows, for each main market, the pre-game probability, the current live probability, and the arithmetic difference. A large shift on a market reflects how significantly the match state has altered the probability distribution.

Win Probability Sparkline

A small line chart showing the history of the home team's win probability from kickoff to the current minute. This history is stored locally in the browser session — it accumulates while the page is open and resets on reload. Sharp upward jumps correspond to home goals; sharp downward drops correspond to away goals.

Live Joint Score PMF Heatmap

The same heatmap as on Page 2, updated in real time. Two visual differences from the pre-game version:

- The cell corresponding to the current live score is outlined in red.
- All cells representing scores impossible from the current state (home goals below the current home score, or away goals below the current away score) carry zero probability and appear dark. The remaining probability is distributed only across reachable final scores.

Next Goal Probabilities

Three values computed from the remaining expected goals λ_{h_rem} and λ_{a_rem} :

Home team scores next: $\lambda_{h_rem} / (\lambda_{h_rem} + \lambda_{a_rem})$ — given that another goal is scored before the final whistle, the probability it goes to the home team.

Away team scores next: $\lambda_{a_rem} / (\lambda_{h_rem} + \lambda_{a_rem})$ — same for the away team.

No more goals: $e^{(-\lambda_{h_rem})} \times e^{(-\lambda_{a_rem})}$ — the joint probability that both Poisson processes produce zero additional goals, i.e., the current score is the final score.

Top 10 Most Likely Final Scores (Live)

The same ranked list as on Page 2, but filtered to exclude scores that are no longer reachable. The current live score is marked with an arrow. As the match progresses and the remaining time shrinks, the top score's probability typically increases because fewer reachable outcomes remain.

Home and Away Marginal Charts

Bar charts showing $P(\text{team scores exactly } k \text{ regulation-time goals})$ for each team, derived from the live PMF in the same way as on Page 2. Only values at or above the team's current score are non-zero.

Scope and Limitations — All Pages

- All probabilities on all three pages represent regulation time (90 minutes plus stoppage time) only. Extra time and penalty shootouts are not included.
- The independence assumption in the Poisson model means the model does not directly account for the correlation introduced by tactical changes after goals. Market reconciliation partially corrects for this.
- Calibration is based on 128 World Cup matches from 2018 and 2022. Out-of-sample performance on the 2026 tournament specifically is unknown at this stage of the tournament.
- Edge estimates are outputs of a probabilistic model. They are not profit guarantees. Market odds move between the time predictions are generated and kickoff. Always verify current odds before acting.
- The WebSocket update speed is subject to network latency and server load. The stated target of under 200 milliseconds applies under normal conditions.

All probabilities represent regulation time (90 minutes + stoppage time) only. Extra time and penalties are excluded. This guide is for informational and educational purposes. Please gamble responsibly.